

Sec-A

1. What is IPsec, and how does it enhance network security?

Ans IPsec (Internet Protocol Security) :- A protocol suite used to secure IP communication by authenticating and encrypting each IP packet in a communication session.

Enhancement to Network Security :-

- Provides Confidentiality, integrity and authentication of data.
- Protects data from tampering, eavesdropping and replay attacks in VPNs and secure communications.

2. What is network security and why is it important?

Ans Network security is the practice of safeguarding a computer network and the data it transmits and stores from unauthorized access, damage, or misuse.

It is important because it protects sensitive information, ensures network functionality and helps businesses comply with regulatory requirements.

Section B (2x7=14)

1. Analyze how the attack was carried out and what security layer was compromised.

Ans TO analyze a cyber attack, consider the following general steps attackers often take:

- (i) Reconnaissance - The attacker gathers information about the target system, such as IP addresses, open ports, and services.
- (ii) Exploitation - Using known vulnerabilities or weak credentials the attacker gains unauthorized access.
- (iii) Privilege Escalation - After initial access they try to gain higher-level privileges within the system.

- (iv) Payload Execution - Malicious software (malware, ransomware) is deployed.
- (v) Covering Tracks - logs and traces are deleted to avoid detection.
- Compromised Security Layer(s) :-
- (i) Application layer - if the attacks exploited software bugs or vulnerabilities in web page apps (eg, SQL, injection).
 - (ii) Network layer - if it involved packet sniffing, spoofing or man-in-the-middle (MitM) attacks.
 - (iii) Transport layer - if protocols like TCP/UDP were manipulated (eg. Session Hijacking)
 - (iv) Data link - if physical access or MAC spoofing was used.

examples: if an attacker used a phishing email to steal login credentials and gain access to sensitive files, the web layer (due to social engineering) and Application layer (due to poor authentication) were compromised.

2. What is IPsec, and how does it enhance network security?

Ans IPsec or Internet Protocol Security is a suite of protocols that enhance network security by providing authentication, encryption and integrity of data transmitted over IP networks. It works by securing individual IP packets, protecting them from unauthorized access, tampering and replay attacks.

It uses two main protocols :-

- (i) Authentication Header (AH) - provides data integrity and authentication.
- (ii) Encapsulating Security Payload (ESP) - provides encryption, integrity and optional authentication.

Here's How IPsec enhance network security-

- (i) Authentication - IPsec verifies the identity of the sender and receiver, ensuring that only authorized parties can communicate.
- (ii) Encryption - it encrypts the data packets, making them unreadable to unauthorized individuals during transmission.
- (iii) Integrity - IPsec ensures that the data remains unchanged during transit by detecting and preventing any modification.
- (iv) Secure VPNs - IPsec is widely used to create virtual private networks (VPNs), enabling secure remote access over public networks.

• Sec - C ($2 \times 11 = 22$)

Q. Discuss the security measures implemented at each layer (Network, Transport, and Application).

Ans. Security of the Network, Transport and application layers focuses on different aspects of data protection.

Here's how network, transport and application layers implement security to protect data and systems from threats.

(i) Network layer security (Layer-3)

Purpose - Protect data as it travels across different networks.

Security Measurements -

- IPsec (Internet Protocol Security) - Encrypts and authenticates IP packets to ensure confidentiality and integrity.
- Firewalls (Layer-3) - Control incoming and outgoing traffic based on IP addresses and protocols.

- Packet Filtering - Filters network traffic based on headers source / destination IP, and ports.
- VPNs (Virtual Private Networks) - Create secure tunnels for data over public networks using encryption protocols like IPsec.

(ii) Transport Layer Security (Layer 4): -

Purpose - Secure end-to-end communication between systems or applications.

Security Measures:-

- TLS / SSL (Transport Layer Security / Secure Socket Layer) - Encrypts data in transit to prevent eavesdropping and tampering (used in HTTPS).
- Secure Ports - Use of well-defined secure ports like 443 for HTTPS, ensuring encrypted communication.
- TCP Wrappers :- Provide access control and logging for services like SSH and FTP.
- DOS Protection - Rate-limiting and SYN cookies help prevent denial-of-service attacks.

(iii) Application Layer Security (Layer 7) -

Purpose - Secure data and user interactions at the application level.

Security Measures:-

- Input Validation & Sanitization - Prevents injection attacks like SQL injection and XSS.
- Authentication & Authorization Mechanisms - Use multi-factor authentication (MFA), session management and role-based access control (RBAC).
- Web Application Firewalls (WAF) - Filters and monitors HTTP traffic to protect against application layer attacks.

- Security Headers (HTTP Headers) - Headers like "Content-Security-Policy" and "X-Frame-Options" helps prevent XSS and clickjacking.

2. Write short notes on:

(a) PGP :- Pretty Good Privacy (PGP) is a data encryption and decryption program that provides cryptographic privacy and authentication for data communication. It is mainly used for securing emails, files and digital signatures.

Key features -

- (i) Hybrid Encryption - Combines symmetric-key encryption (for speed) with public-key encryption (for secure key exchange).
- (ii) Data Confidentiality - Encrypts message content so only the intended recipient can decrypt and read it.
- (iii) Authentication - Uses digital signature to verify the sender's identity.
- (iv) Integrity - Ensures the message has not been altered using hashing algorithms like SHA.
- (v) Compression - Reduces message size before encryption, increasing efficiency and security.

• Applications:-

- ↳ Secure email communication
- ↳ File encryption
- ↳ Code signing & software integrity.

(b) SET (Secure Electronic Transaction) -

SET is a security protocol developed by Visa and MasterCard to secure online credit card

Transaction. It employs digital certificates, encryption and

digital signatures to ensure the secure exchange of payment information between parties.

Key Features:

- (i) Cardholder Authentication - Confirms the ~~buy~~ buyer's identity using digital certificates.
- (ii) Merchant Authentication - verifies the legitimacy of the merchant to the customer.
- (iii) Data Confidentiality - Encrypts credit card information and payment details so they remain private during transaction.
- (iv) Transaction Integrity - Uses digital signatures to ensure that payment information is not altered during transit.
- (v) Dual Signatures - Separates payment and order info to protect consumer and merchant privacy while linking both securely.

Applications —

- ↳ Online Shopping
 - ↳ E-commerce Platforms
 - ↳ Secure B2C transactions.
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